Data Science and AI for industry

Term Project Presentation

Marta Castellano
Department of Statistics and Operations Research
Universitat Politècnica de Catalunya
Barcelona, Spain
marta.castellano@upc.edu



UNIVERSITAT POLITÈCNICA
DE CATALUNYA
BARCELONATECH

Term project presentation

Goal: Respond to a CLEAR question (the goal is not to explore data).

- Option 1: Road accidents in Catalunya
- Option 2: propose a question and a dataset
- Option 3: propose a question and create a dataset

Option 1: Road accidents in Catalunya

- **The question:** Did the removal of highway tolls in Catalonia on **1 September 2021** change the characteristics of severe traffic accidents?

- **Data:** Traffic accidents involving **death and/or severe injury in Catalonia, 2010–2024**.

- **Your challenge:**
  - **Explore** how accident characteristics changed **before vs. after** toll removal.
  - **Build a classifier** to distinguish accidents occurring before vs. after the change.
  - **Interpret the model** to identify which accident characteristics drive the distinction.

- **Critical points:** class imbalance, changes over time, derived features, validation strategy, appropriate performance metrics.



Term project presentation



Term project presentation

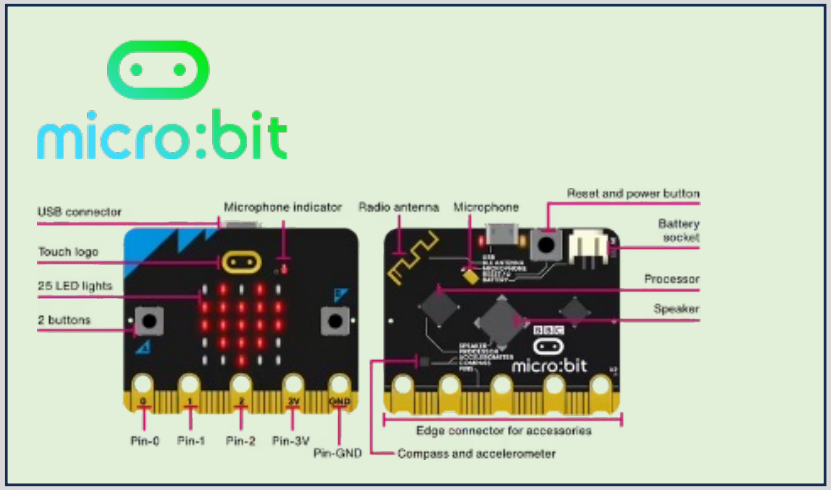
Option 2: Propose a question and a dataset

- Databases
  - https://ourworldindata.org/
  - https://ieee-dataport.org/datasets
  - https://archive.ics.uci.edu/ UCL machine learning repository
  - https://archive.ics.uci.edu/
  - https://lib.stat.cmu.edu/datasets/
  - https://datasetsearch.research.google.com/
  - https://data.gov/
  - https://github.com/search?q=dataset
  - https://data.nasa.gov/

Term project presentation

Option 3: Propose a question and create a dataset

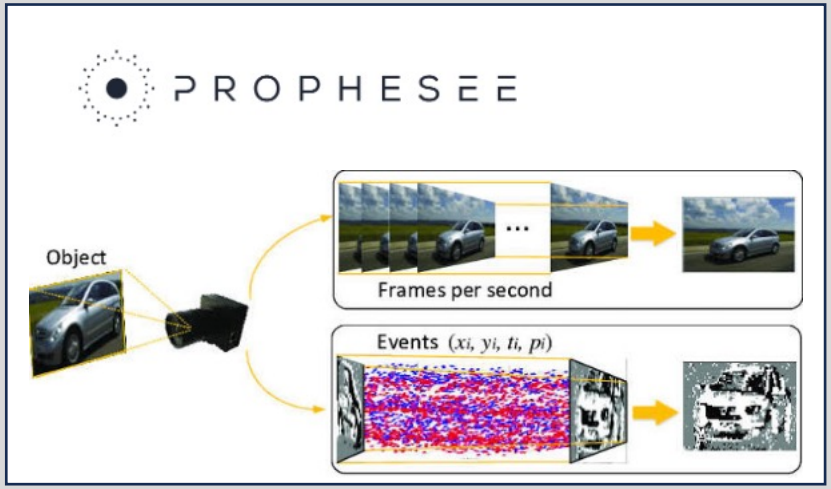
Microcontroller with sensors



Bio signals



Neuromorphic camera



Term project presentation

The dataset must:

- Respond to a CLEAR question (the goal is not to explore data).
- Data complexity: (at least 2)
  - High dimensionality (not too much either…)
  - Temporal component
  - Class imbalance
  - Multiple sources
  - Repeated measures
- Minimum size requirements:
  - Tabular data: $5000 - 50000$ observations and $\geq 8 - 10$ variables (mixture of numeric and categorical)
  - Images, text or time series: minimum 100
- Data quality requirements:
  - Raw data should not be ready for analysis (i.e Kaggle)
  - At least 3 of the following is needed: missing-value strategy, feature engineering, merging tables, aggregation

Term project presentation

Events and deadlines

- **Monday 5 Oct:** submit groups (2 to 3 students) and option. In case a team wants to use its own dataset and objectives (option 2 or 3), it has to bee agreed with the professors beforehand.

- **October:** Review meeting with professors during October. Professors will send slots to have a Google Meet meeting to review project progress.

- **Friday 27 Nov (23:59):** term project submission in the corresponding Atenea task.